Music Minimal Skip Optimizer

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Problem Statement

Skipping Songs

Skip behavior is core engagement sign for music platforms.

Skips can indicate:

- Shorter listening sessions
- Lower satisfaction
- Reduced retention

Objective

Given a fixed set of candidate tracks, find the best permutation to minimize cumulative predicted skip probability

Impact

Ordering is a real product decision (Daily Mix, Radio, Discover Weekly)

Better ordering improves listening flow without changing the underlying songs.

Pipeline Overview

1. Input Data

Spotify session events + track/context features

2. Skip Modeling

Train transformer model and build a skip scorer

3. Model Handoff

Use trained skip scorer to answer “given an ordering, what is # expected skips?”

4. Ordering Optimization

Optimize ordering for the least expected total skips using Birkhoff Polytope

5. Final Comparison

Identity vs Greedy vs Birkhoff Optimization

6. Headline Result

Birkhoff: 7.5181 -> 5.1244 expected skips

Data

Spotify Sequential Skip Prediction (WSDM Cup 2018)

Data Updates

- Merged Spotify session logs + track metadata/audio features
- 167,880 merged rows
- 8,000 train / 2,000 test sessions

Feature Engineering

- Where the user is in the session and time context (*session_progress*, day/time/weekend)
- How smooth the transition is (*delta_**, acoustic similarity)
- Cumulative user behavior so far (*cum_skips*, *skip_rate_so_far*, pauses, tempo deviation)

Why it's suitable

- Moved from mostly heuristic signals to a context-aware feature space

Skip Model: Transformer Methodology

Goal: Predict skip vs no-skip for the current track event using prior session history and current-event context.

Original idea: Transformer model with input of compound tokens which consist of multiple features, and output of two heads of SKIP or SONG (no skip).

The output next token is decoded by comparing to a threshold, which predicts a skip.

Skip F1 Score: 0.63

Skip Model: Further Optimizations

Variation A

First direct-skip emphasis: reduce song loss weight to 0, emphasis on skip loss.

skip_f1 = 0.6878

Variation B

Direct per-track skip target: change input to history up to current song, label becomes skip/no skip using BCE (from joint token for song/skip).

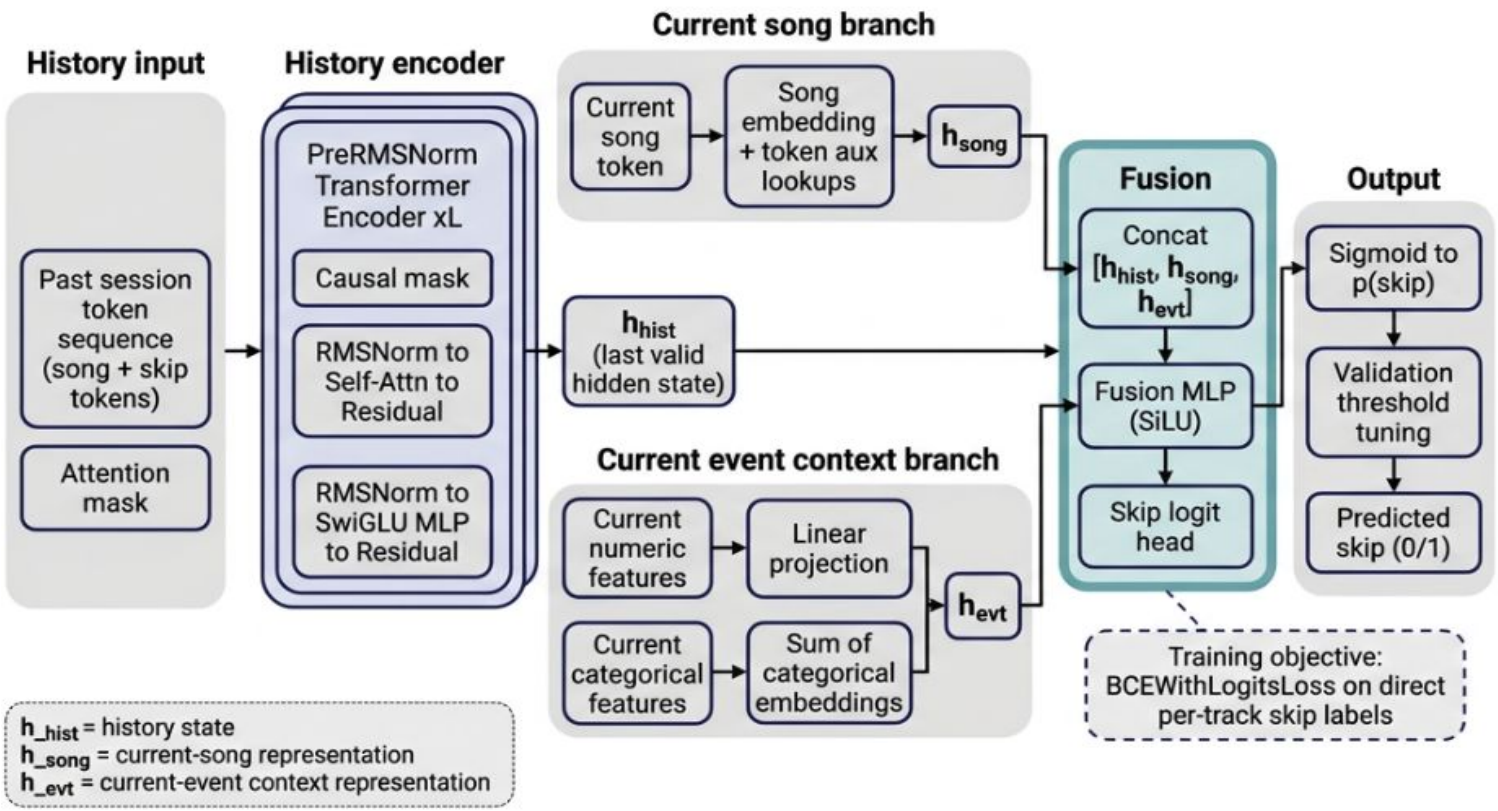
skip_f1 = 0.7066

Variation C

Direct skip + current event context fusion: introduce fusion architecture by concatenating embeddings for history, current song, and context.

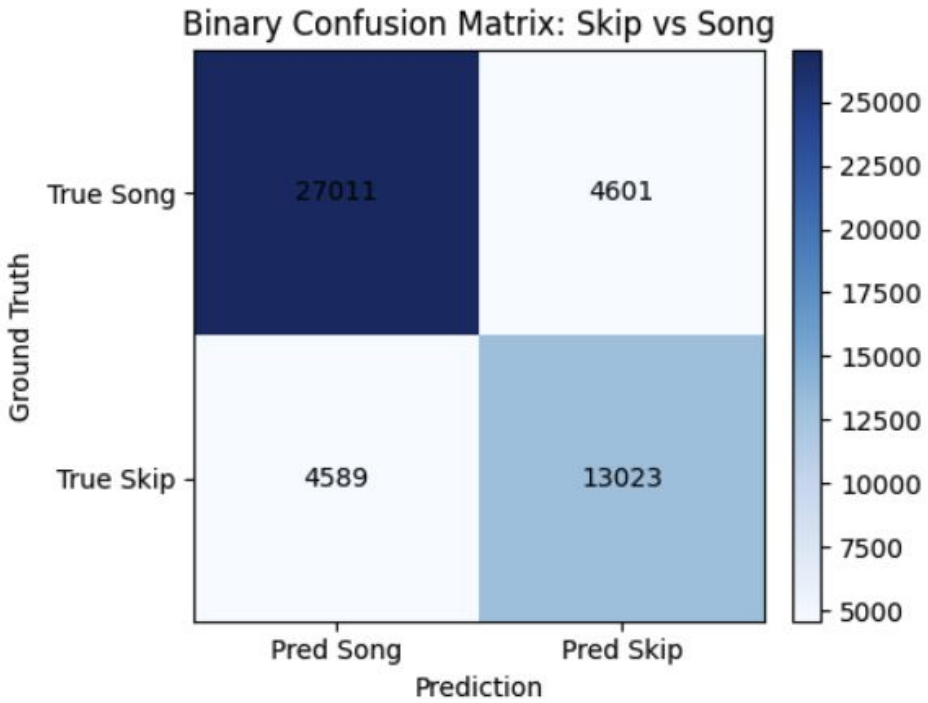
skip_f1 = 0.8009

Skip Model: Transformer Architecture (Variant C)



Skip Model: Transformer Model (C) Results

- The skip precision is 75%, and the recall is 86% (vs. 73% and 63% for the baseline probability model).
- Skip F1: 0.8009
- Optimized threshold for skipping is 0.43.
- Overall solid and robust performance.



Ordering Model: Birkhoff Polytope

- **Goal:** Given *n = 20* songs and the black-box skip predictor, find the permutation of those *n* songs that minimizes the expected # of skips, without enumerating all *n!* Permutations.

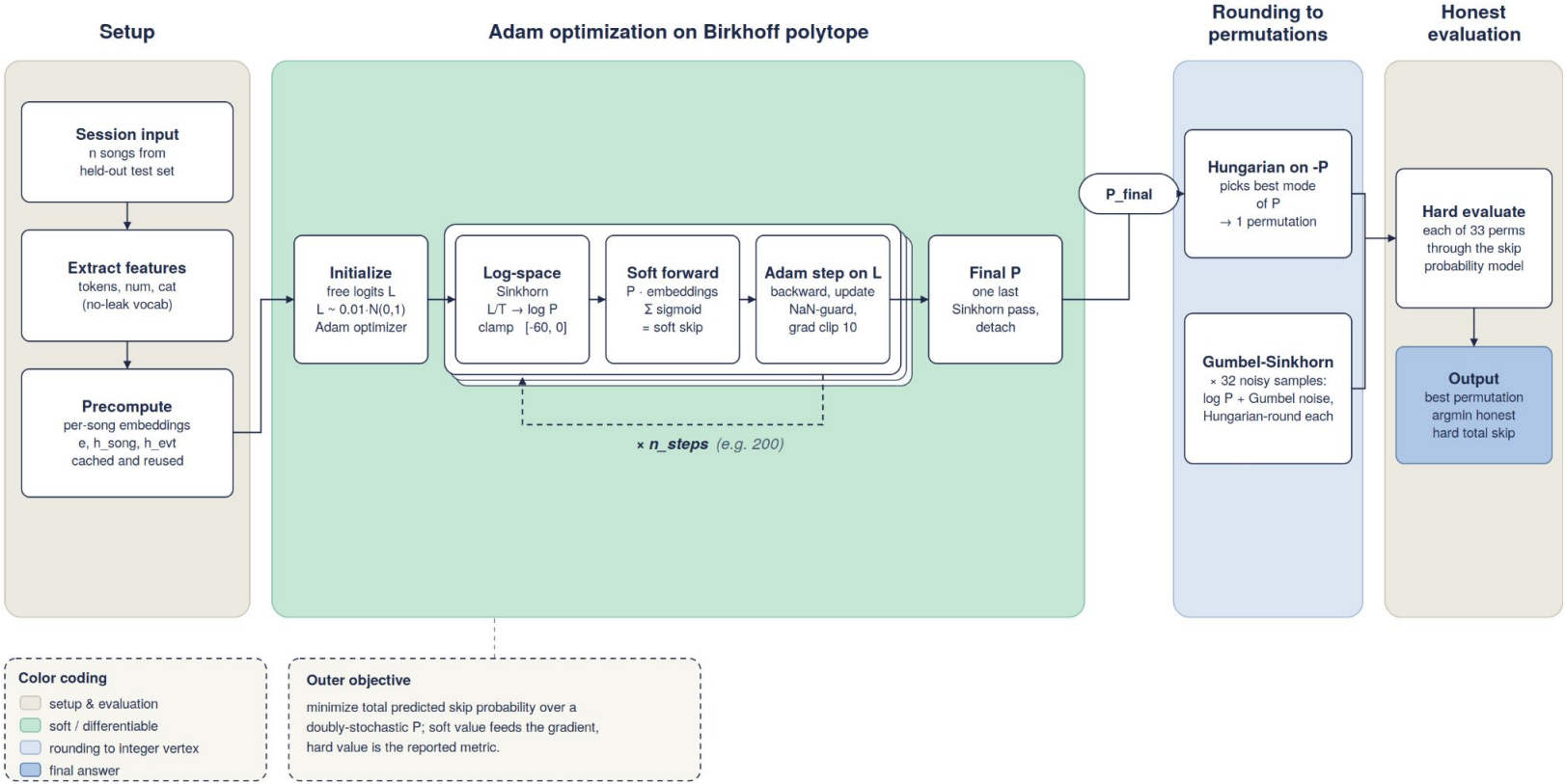
- **Birkhoff Polytope**
  - ***Def:*** The set of **doubly stochastic matrices** (rows sum to 1, columns sum to 1, and entries are in [0, 1]) is the **convex hull** of permutation matrices (corner vertices).

  - Thus, we optimize over the polytope (through Adam), get close to a vertex, then snap to that nearest vertex (and compare it with other close vertices) to obtain final ordering!

- **Ex: Permutation Matrix**

	song0	**song1**	**song2**	
pos0	0	1	0	← position 0 gets song 1
pos1	0	0	1	← position 1 gets song 2
pos2	1	0	0	← position 2 gets song 0

Ordering Model: Birkhoff Polytope



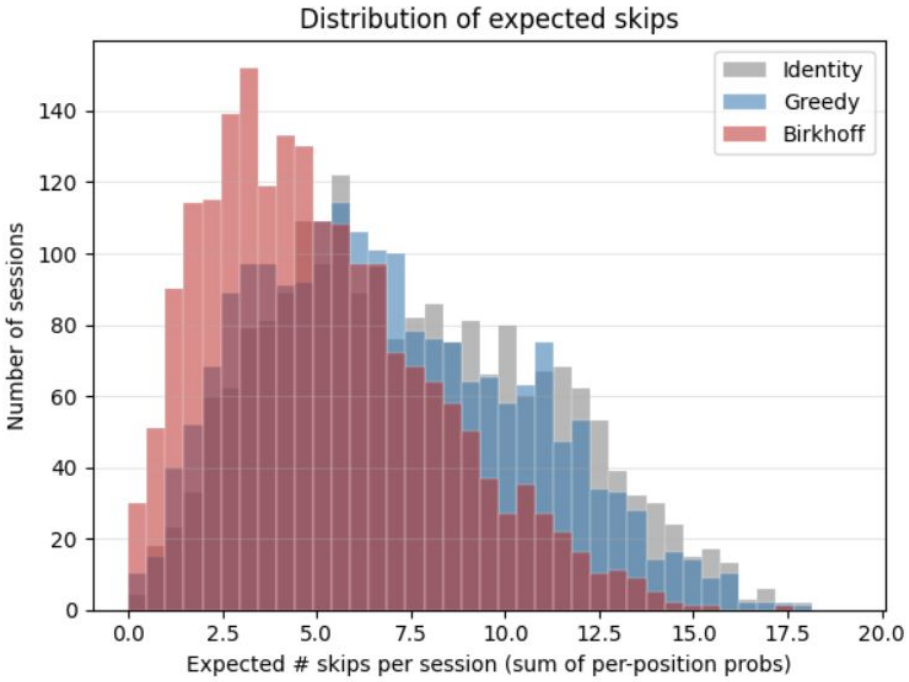
Ordering Results: Expected Skip Summary

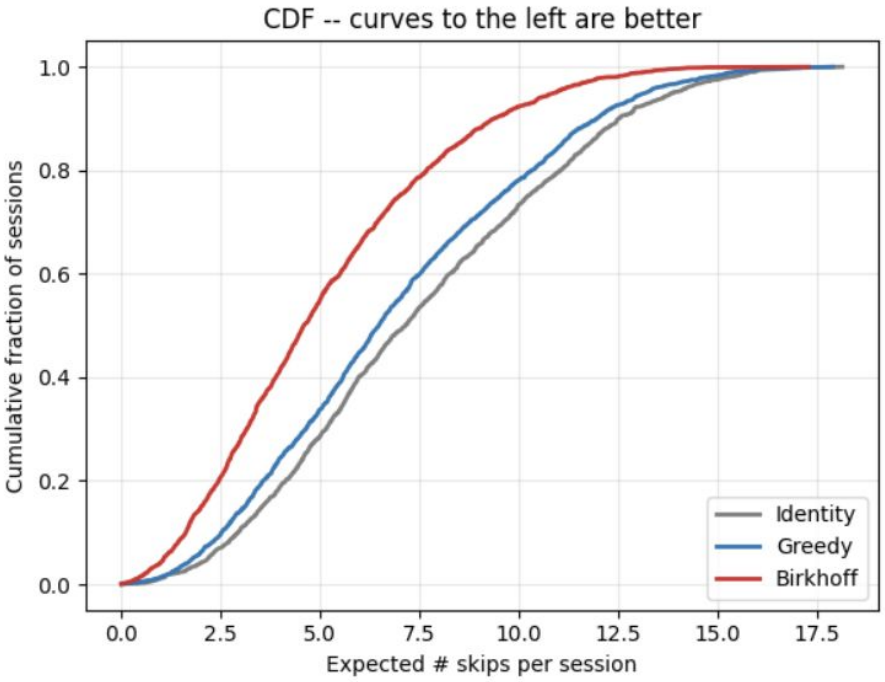
Method	Mean Skips [95% CI (n = 2000)]	% Identity Reduction
Identity	**7.5181** [7.3578, 7.6784]	0.00%
Greedy	**6.9298** [6.7738, 7.0858]	7.55%
Birkhoff	**5.1244** [4.9926, 5.2562]	32.98%

Birkhoff < Identity : 1954/2000 (97.7%) sessions
Birkhoff < Greedy : 1860/2000 (93%) sessions

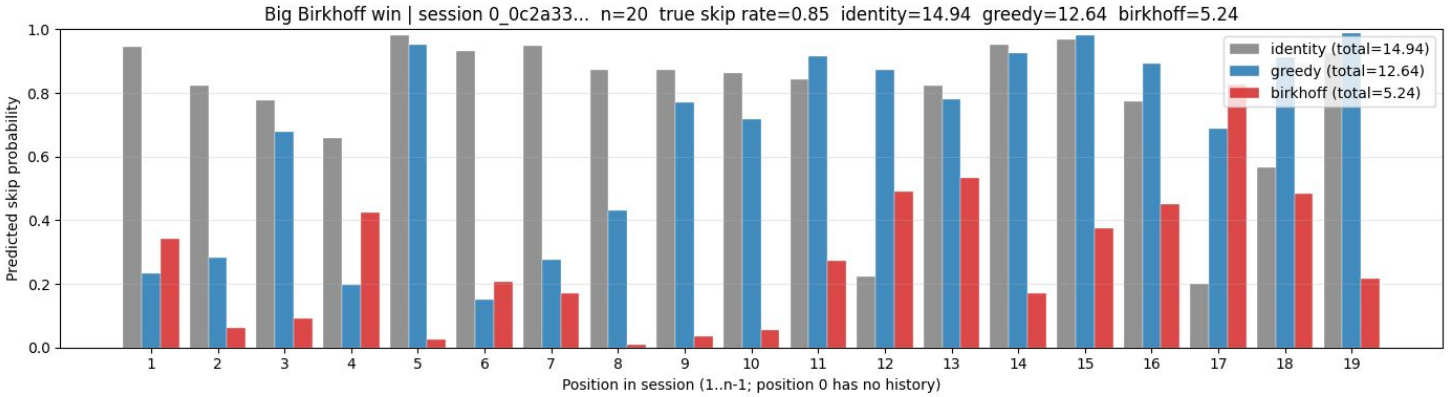
In a 20 song session, Birkhoff ordering reduces the average expected # skips by ~2.39, which is a 33% reduction!

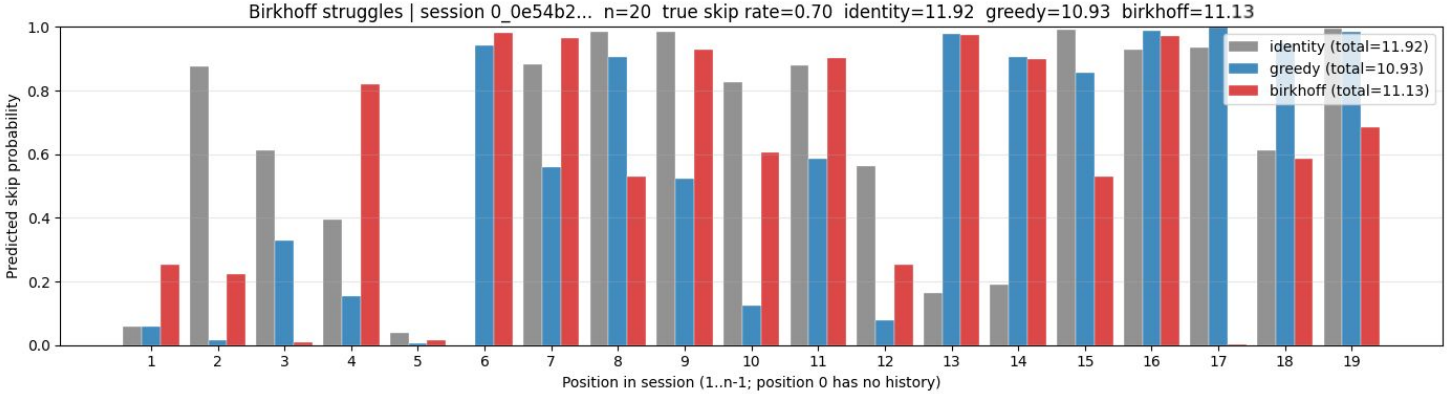
Ordering Results: Expected Skip Distribution





Ordering Results: Birkhoff Wins & Losses





Future Work

- Scale the model using more data -> can improve F1 score of transformer
  - More sessions available in full dataset (currently unavailable as Spotify removed)

- Work with longer context windows and explore cross session history for personalization on a user level

- Expand objectives: product teams might care about artist-fairness and diversity -> can minimize skips subject to other constraints

Thank You!